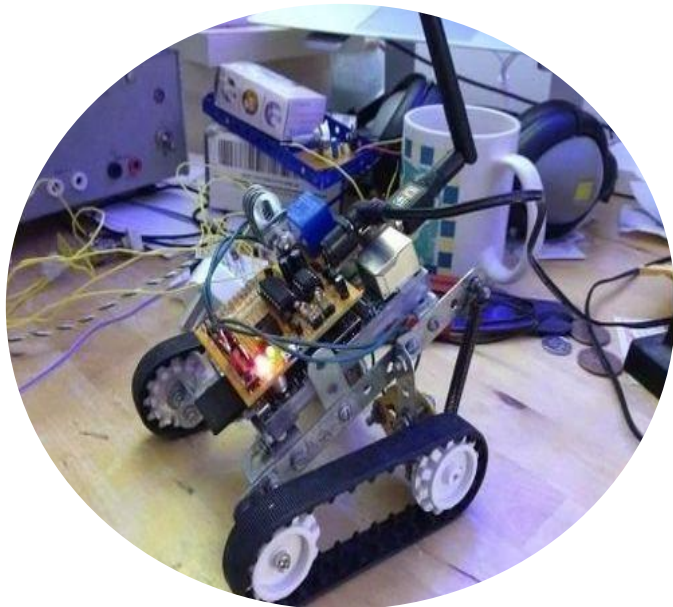
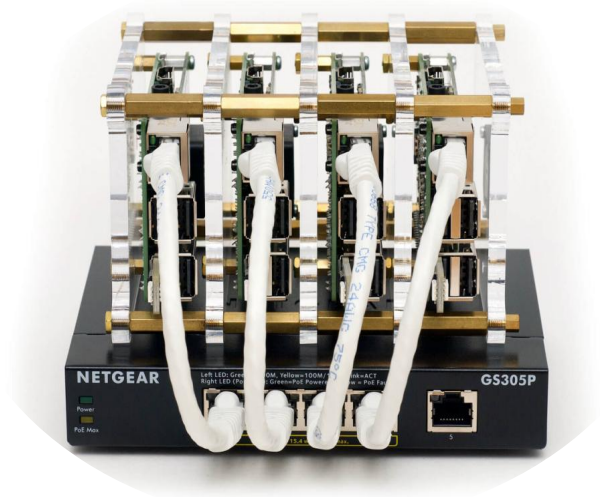




微机原理与微系统实验-树莓派



主讲老师：王小静
办公地点：慧园2栋411
open office hours：周三14:00-16:00

提交截图：通过自己的笔记本电脑，远程登录自己的树莓派，之后打开BB的界面。

Submit a Screenshot: Remotely log in to your own Raspberry Pi via your laptop, and then open the website of your course BB.

作答

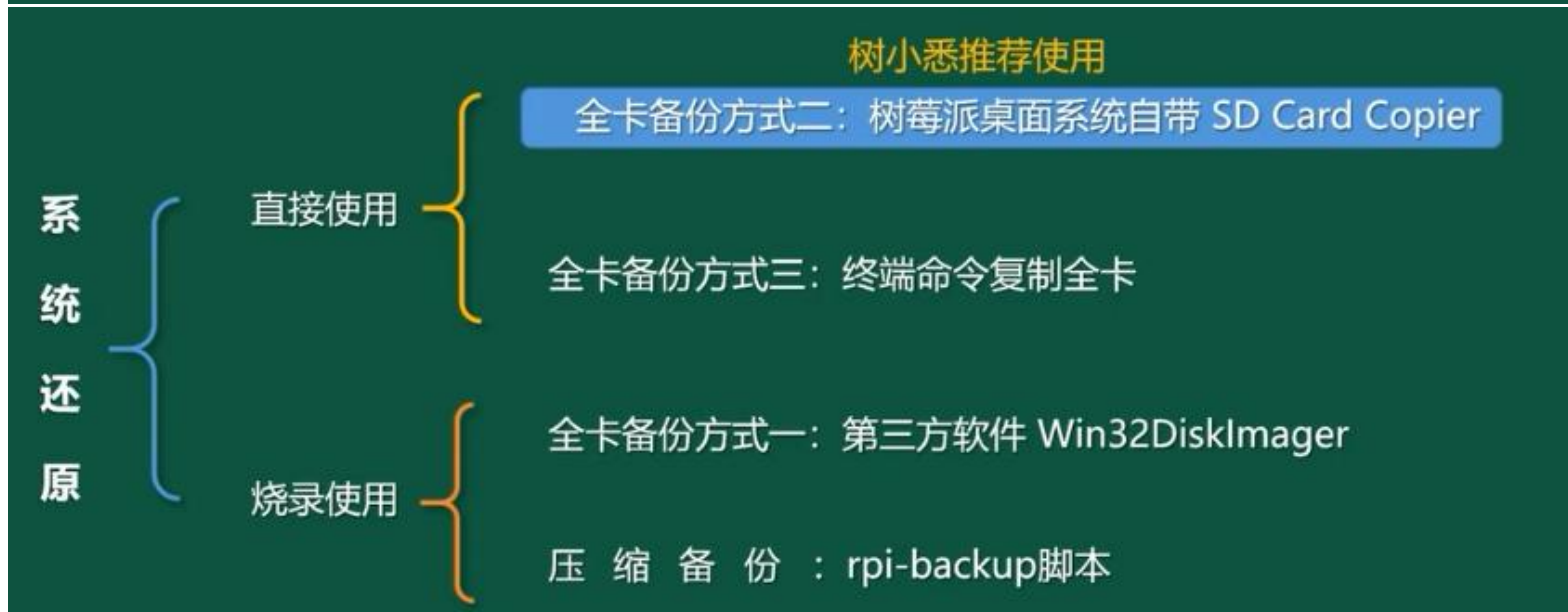
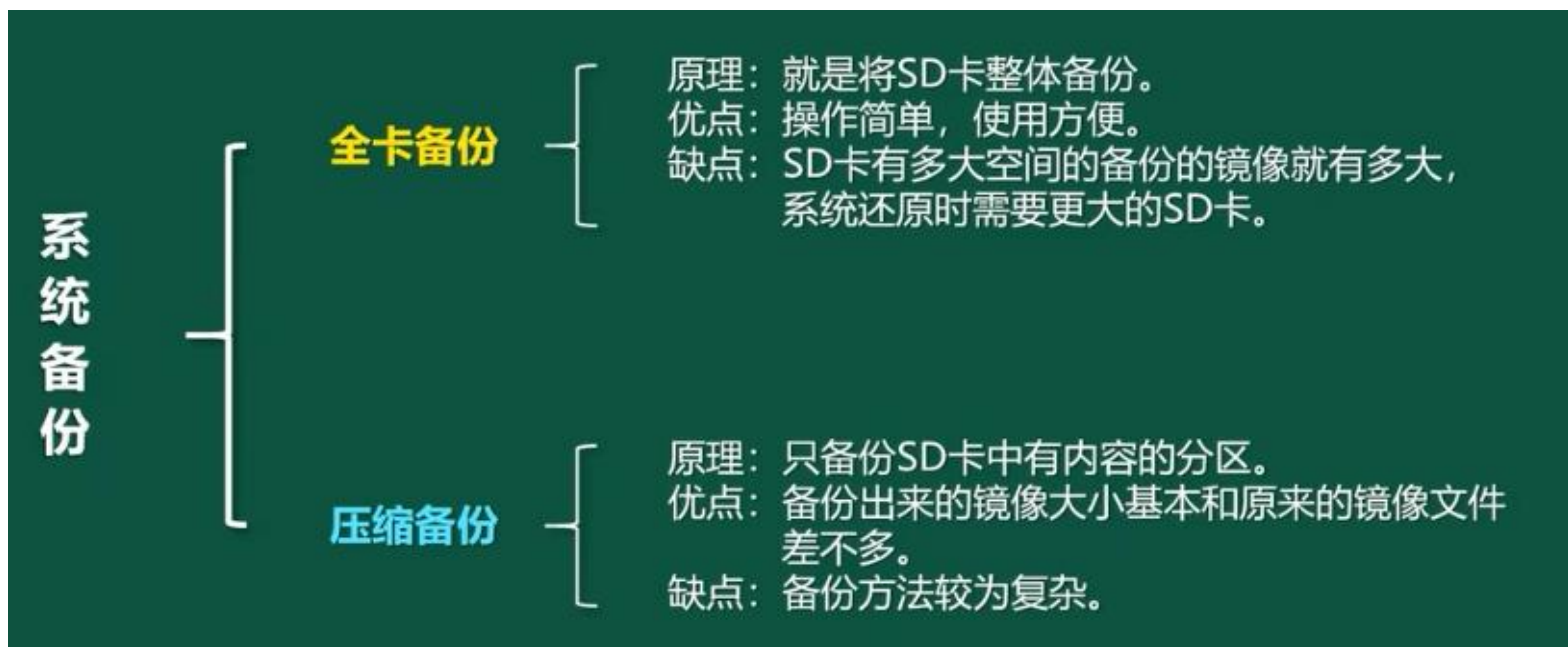
Class 2: 编程基础及双色灯实验 (Preparatory Knowledge and Dual-Color LED Experiment)

树莓派系统的备份与还原

(Backup and Restore of Raspberry Pi System)

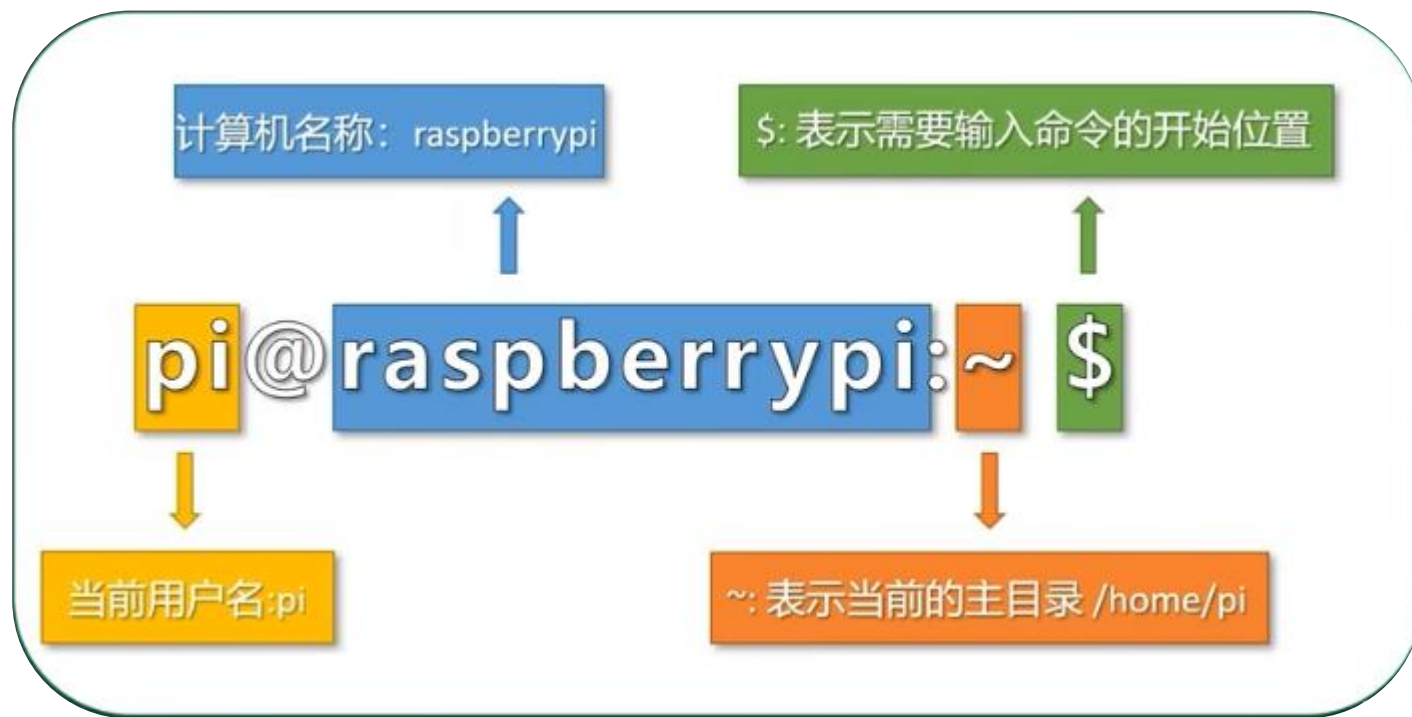
- `sudo rm -rf /` #删除树莓派的根目录，将系统进行格式化删除，没得补救
 - **在没有备份之前不要随便使用不懂的命令**
- Do not run any unfamiliar commands until you have a backup.**





https://www.bilibili.com/video/BV16U4y1879Q?p=10&spm_id_from=pageDriver&vd_source=dbcd5054bc4c43b1f2ca03babf12dc07

Linux常用终端命令 (Common Terminal Commands)



身份	普通用户（ pi 或其他自定义账号 ）	超级用户（ root ）
用户 ID	UID ≥ 1000	UID = 0
提示符	\$	#
系统文件写入	只读，需 sudo 才能改 /etc、 /usr 等	直接读写，无限制
软件安装	sudo apt install 需再输密码	apt install 一步到位
硬件端口控制	要加入 gpio 组或 sudo	直接读写 /dev/gpiomem、 /dev/i2c-*
服务启停	sudo systemctl start xxx	systemctl start xxx
风险等级	误操作影响有限	一句 rm -rf /* 系统直接消失
登录方式	默认允许桌面/SSH 登录	默认禁用，需手动设密码或 sudo su
桌面环境	完整图形界面，自动登录	无桌面、无主题，纯命令行
推荐用法	日常开发、调试、跑程序	仅系统救援、批量脚本、制作镜像

一句话总结：普通用户是“带安全带的驾驶员”；超级用户是“拆掉所有安全线的赛车手”，速度虽快，翻车也更惨。树莓派官方推荐能不用 root 就不用 root，日常在普通用户下加 sudo 是最稳妥的做法。



sudo : SuperUser Do

以超级用户身份去执行

A terminal window on a Raspberry Pi showing the following commands and output:

```
pi@raspberrypi:~ $ sudo su
root@raspberrypi:/home/pi# su pi
pi@raspberrypi:~ $
```

A blue arrow labeled "普通用户pi" points to the first prompt. A yellow arrow labeled "超级用户root" points to the second prompt.

分类	命令	含义
目录切换	pwd	# 显示当前所在目录
	cd ~	#切换到主目录 (/home/pi) .~可以不写
	cd dir	#切换到指定目录，dir表示文件路径
	cd ..	#切换到上一级目录
文件展示	ls	#展示当前目录下所有的文件和文件夹（不包含隐藏文件）
	ls -a	#展示当前目录下所有的文件和文件夹（包含隐藏文件）
创建	touch file	#创建文件file
	mkdir dir	#创建目录dir
查看	cat file	#查看文件file内容
	more file	#查看文件file内容
	head file	#查看文件file前10行
	tail file	#查看文件file后10行
删除	rm file	#删除文件file
	rm -r dir	#删除目录dir
重命名或移动	mv file 1 file2	#若file2不存在，则将file1改名为file2；若file2存在，则file1覆盖掉原file2，并更名为file2
	mv file dir	#将文件file移动到dir目录中

基础--编程语言 (Programming Language)



官网已经内嵌了python2和python3，需要注意一下使用的python版本

```
pi@raspberrypi:~ $ python
Python 2.7.16 (default, Oct 10 2019, 22:02:15)
[GCC 8.3.0] on linux2
Type "help", "copyright", "credits" or "license" for more information.
>>> print "hello world"
hello world
>>> exit()
pi@raspberrypi:~ $
```

```
pi@raspberrypi:~ $ python3
Python 3.7.3 (default, Jan 22 2021, 20:04:44)
[GCC 8.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> print "hello world"
  File "<stdin>", line 1
    print "hello world"
        ^
SyntaxError: Missing parentheses in call to 'print'. Did you mean print("hello world")?
>>> print("hello world")
hello world
>>> exit()
pi@raspberrypi:~ $
```

```
pi@raspberrypi:~ $ python --version
Python 2.7.16
pi@raspberrypi:~ $ python3 --version
Python 3.7.3
pi@raspberrypi:~ $
```

Python2 安装命令

```
pip install pandas # 这里以pandas库为例
```

Python3 安装命令

```
pip3 install pandas # 这里以pandas库为例
```

想用C/C++编程--安装Wiring Pi库 (WiringPi是控制树莓派GPIO的C代码库)



```
pi@raspberrypi:~ $ cd /tmp
pi@raspberrypi:/tmp $ wget https://project-downloads.drogon.net/wiringpi-latest.deb
--2021-09-16 15:59:24-- https://project-downloads.drogon.net/wiringpi-latest.deb
正在解析主机 project-downloads.drogon.net (project-downloads.drogon.net)... 188.246.205.22
, 2a03:9800:10:7b::2
正在连接 project-downloads.drogon.net (project-downloads.drogon.net)|188.246.205.22|:443..
. 已连接。
已发出 HTTP 请求，正在等待回应... 200 OK
长度：52260 (51K) [application/x-debian-package]
正在保存至：“wiringpi-latest.deb”

wiringpi-latest.deb 100%[=====>] 51.04K 93.5KB/s 用时 0.5s

2021-09-16 15:59:26 (93.5 KB/s) - 已保存 “wiringpi-latest.deb” [52260/52260]

pi@raspberrypi:/tmp $ sudo dpkg -i wiringpi-latest.deb
```

安装完毕后，可以使用如下命令测试是否成功安装。

```
pi@raspberrypi:~ $ gpio -v
gpio version: 2.52
Copyright (c) 2012-2018 Gordon Henderson
This is free software with ABSOLUTELY NO WARRANTY.
For details type: gpio -warranty

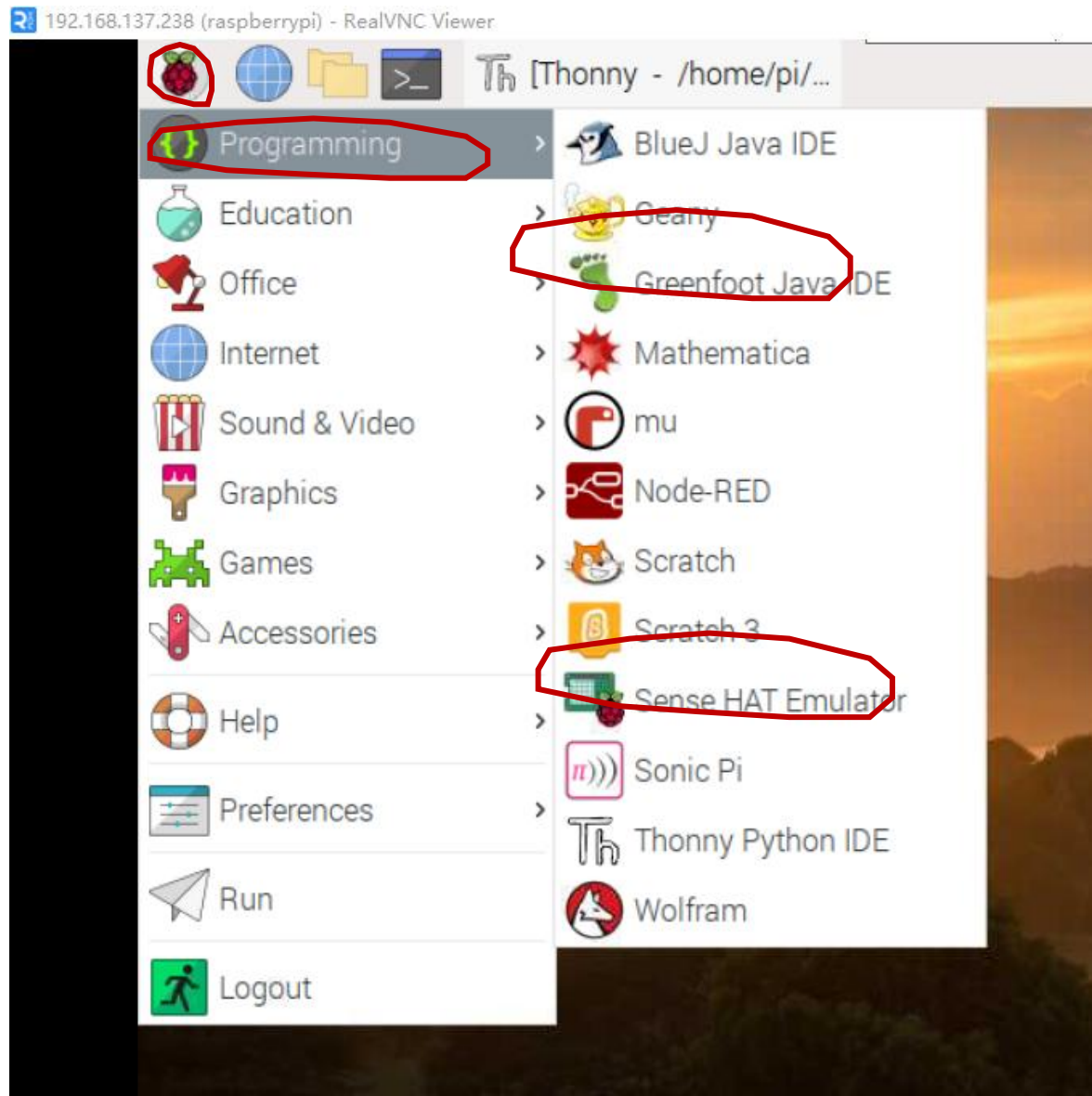
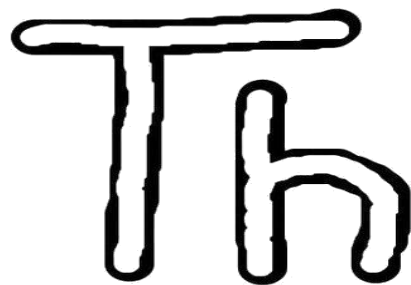
Raspberry Pi Details:
Type: Pi 4B, Revision: 04, Memory: 4096MB, Maker: Sony
* Device tree is enabled.
*--> Raspberry Pi 4 Model B Rev 1.4
* This Raspberry Pi supports user-level GPIO access.

pi@raspberrypi:~ $ gpio readall
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| BCM | wPi |   Name   | Mode | V | Physical | V | Mode | Name      | wPi | BCM |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|  0  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   |  3  |  0  |
|  1  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       |  1  |  1  |
|  2  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   |  4  |  2  |
|  3  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       |  5  |  3  |
|  4  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   |  6  |  4  |
|  5  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       |  7  |  5  |
|  6  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   |  8  |  6  |
|  7  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       |  9  |  7  |
|  8  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 10  |  8  |
|  9  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 11  |  9  |
| 10  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 12  | 10  |
| 11  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 13  | 11  |
| 12  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 14  | 12  |
| 13  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 15  | 13  |
| 14  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 16  | 14  |
| 15  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 17  | 15  |
| 16  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 18  | 16  |
| 17  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 19  | 17  |
| 18  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 20  | 18  |
| 19  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 21  | 19  |
| 20  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 22  | 20  |
| 21  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 23  | 21  |
| 22  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 24  | 22  |
| 23  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 25  | 23  |
| 24  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 26  | 24  |
| 25  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 27  | 25  |
| 26  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 28  | 26  |
| 27  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 29  | 27  |
| 28  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 30  | 28  |
| 29  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 31  | 29  |
| 30  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 32  | 30  |
| 31  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 33  | 31  |
| 32  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 34  | 32  |
| 33  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 35  | 33  |
| 34  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 36  | 34  |
| 35  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 37  | 35  |
| 36  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 38  | 36  |
| 37  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 39  | 37  |
| 38  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 40  | 38  |
| 39  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 41  | 39  |
| 40  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 42  | 40  |
| 41  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 43  | 41  |
| 42  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 44  | 42  |
| 43  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 45  | 43  |
| 44  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 46  | 44  |
| 45  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 47  | 45  |
| 46  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 48  | 46  |
| 47  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 49  | 47  |
| 48  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 50  | 48  |
| 49  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 51  | 49  |
| 50  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 52  | 50  |
| 51  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 53  | 51  |
| 52  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 54  | 52  |
| 53  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 55  | 53  |
| 54  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 56  | 54  |
| 55  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 57  | 55  |
| 56  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 58  | 56  |
| 57  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 59  | 57  |
| 58  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 60  | 58  |
| 59  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 61  | 59  |
| 60  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 62  | 60  |
| 61  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 63  | 61  |
| 62  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 64  | 62  |
| 63  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 65  | 63  |
| 64  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 66  | 64  |
| 65  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 67  | 65  |
| 66  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 68  | 66  |
| 67  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 69  | 67  |
| 68  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 70  | 68  |
| 69  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 71  | 69  |
| 70  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 72  | 70  |
| 71  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 73  | 71  |
| 72  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 74  | 72  |
| 73  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 75  | 73  |
| 74  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 76  | 74  |
| 75  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 77  | 75  |
| 76  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 78  | 76  |
| 77  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 79  | 77  |
| 78  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 80  | 78  |
| 79  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 81  | 79  |
| 80  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 82  | 80  |
| 81  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 83  | 81  |
| 82  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 84  | 82  |
| 83  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 85  | 83  |
| 84  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 86  | 84  |
| 85  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 87  | 85  |
| 86  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 88  | 86  |
| 87  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 89  | 87  |
| 88  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 90  | 88  |
| 89  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 91  | 89  |
| 90  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 92  | 90  |
| 91  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 93  | 91  |
| 92  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 94  | 92  |
| 93  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 95  | 93  |
| 94  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 96  | 94  |
| 95  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 97  | 95  |
| 96  |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 98  | 96  |
| 97  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 99  | 97  |
| 98  |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 100 | 98  |
| 99  |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 101 | 99  |
| 100 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 102 | 100 |
| 101 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 103 | 101 |
| 102 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 104 | 102 |
| 103 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 105 | 103 |
| 104 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 106 | 104 |
| 105 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 107 | 105 |
| 106 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 108 | 106 |
| 107 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 109 | 107 |
| 108 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 110 | 108 |
| 109 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 111 | 109 |
| 110 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 112 | 110 |
| 111 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 113 | 111 |
| 112 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 114 | 112 |
| 113 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 115 | 113 |
| 114 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 116 | 114 |
| 115 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 117 | 115 |
| 116 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 118 | 116 |
| 117 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 119 | 117 |
| 118 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 120 | 118 |
| 119 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 121 | 119 |
| 120 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 122 | 120 |
| 121 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 123 | 121 |
| 122 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 124 | 122 |
| 123 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 125 | 123 |
| 124 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 126 | 124 |
| 125 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 127 | 125 |
| 126 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 128 | 126 |
| 127 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 129 | 127 |
| 128 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 130 | 128 |
| 129 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 131 | 129 |
| 130 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 132 | 130 |
| 131 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 133 | 131 |
| 132 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 134 | 132 |
| 133 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 135 | 133 |
| 134 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 136 | 134 |
| 135 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 137 | 135 |
| 136 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 138 | 136 |
| 137 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 139 | 137 |
| 138 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 140 | 138 |
| 139 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 141 | 139 |
| 140 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 142 | 140 |
| 141 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 143 | 141 |
| 142 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 144 | 142 |
| 143 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 145 | 143 |
| 144 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 146 | 144 |
| 145 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 147 | 145 |
| 146 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 148 | 146 |
| 147 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 149 | 147 |
| 148 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 150 | 148 |
| 149 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 151 | 149 |
| 150 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 152 | 150 |
| 151 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 153 | 151 |
| 152 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 154 | 152 |
| 153 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 155 | 153 |
| 154 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 156 | 154 |
| 155 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 157 | 155 |
| 156 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 158 | 156 |
| 157 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 159 | 157 |
| 158 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 160 | 158 |
| 159 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 161 | 159 |
| 160 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 162 | 160 |
| 161 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 163 | 161 |
| 162 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 164 | 162 |
| 163 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 165 | 163 |
| 164 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 166 | 164 |
| 165 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 167 | 165 |
| 166 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 168 | 166 |
| 167 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 169 | 167 |
| 168 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 170 | 168 |
| 169 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 171 | 169 |
| 170 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 172 | 170 |
| 171 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 173 | 171 |
| 172 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 174 | 172 |
| 173 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 175 | 173 |
| 174 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 176 | 174 |
| 175 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 177 | 175 |
| 176 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 178 | 176 |
| 177 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 179 | 177 |
| 178 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 180 | 178 |
| 179 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 181 | 179 |
| 180 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 182 | 180 |
| 181 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 183 | 181 |
| 182 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 184 | 182 |
| 183 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 185 | 183 |
| 184 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 186 | 184 |
| 185 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 187 | 185 |
| 186 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 188 | 186 |
| 187 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 189 | 187 |
| 188 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 190 | 188 |
| 189 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 191 | 189 |
| 190 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 192 | 190 |
| 191 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 193 | 191 |
| 192 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 194 | 192 |
| 193 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 195 | 193 |
| 194 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 196 | 194 |
| 195 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 197 | 195 |
| 196 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 198 | 196 |
| 197 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 199 | 197 |
| 198 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 200 | 198 |
| 199 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 201 | 199 |
| 200 |  0  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 202 | 200 |
| 201 |  1  |   GND    |  0    |  0 |      0   |  0 |  0    |  GND       | 203 | 201 |
| 202 |  4  |   VCC    |  0    |  1 |      1   |  1 |  0    | 3V3 (L)   | 204 | 202 |
| 203 |  1  |   GND    |  0    |  0 |      0   | 
```

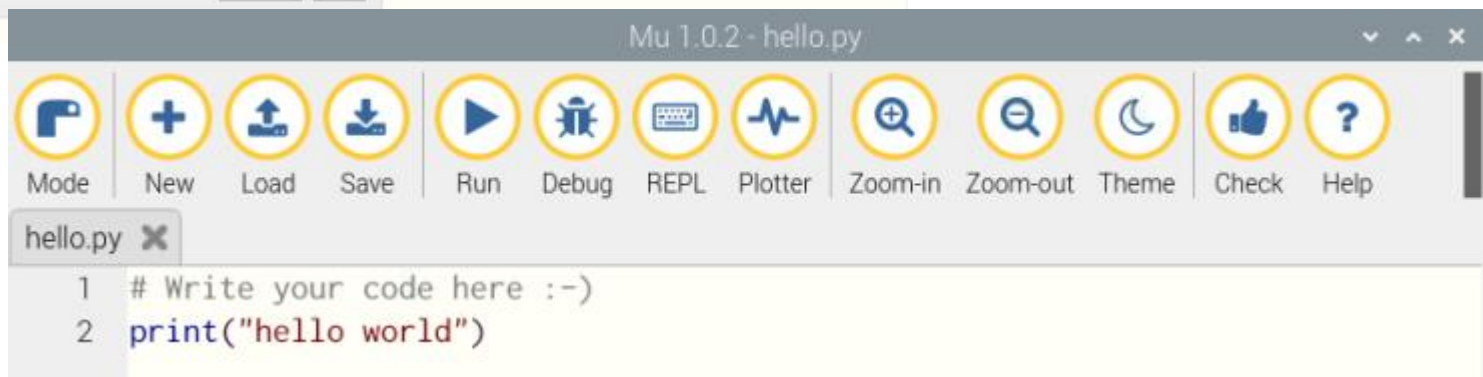
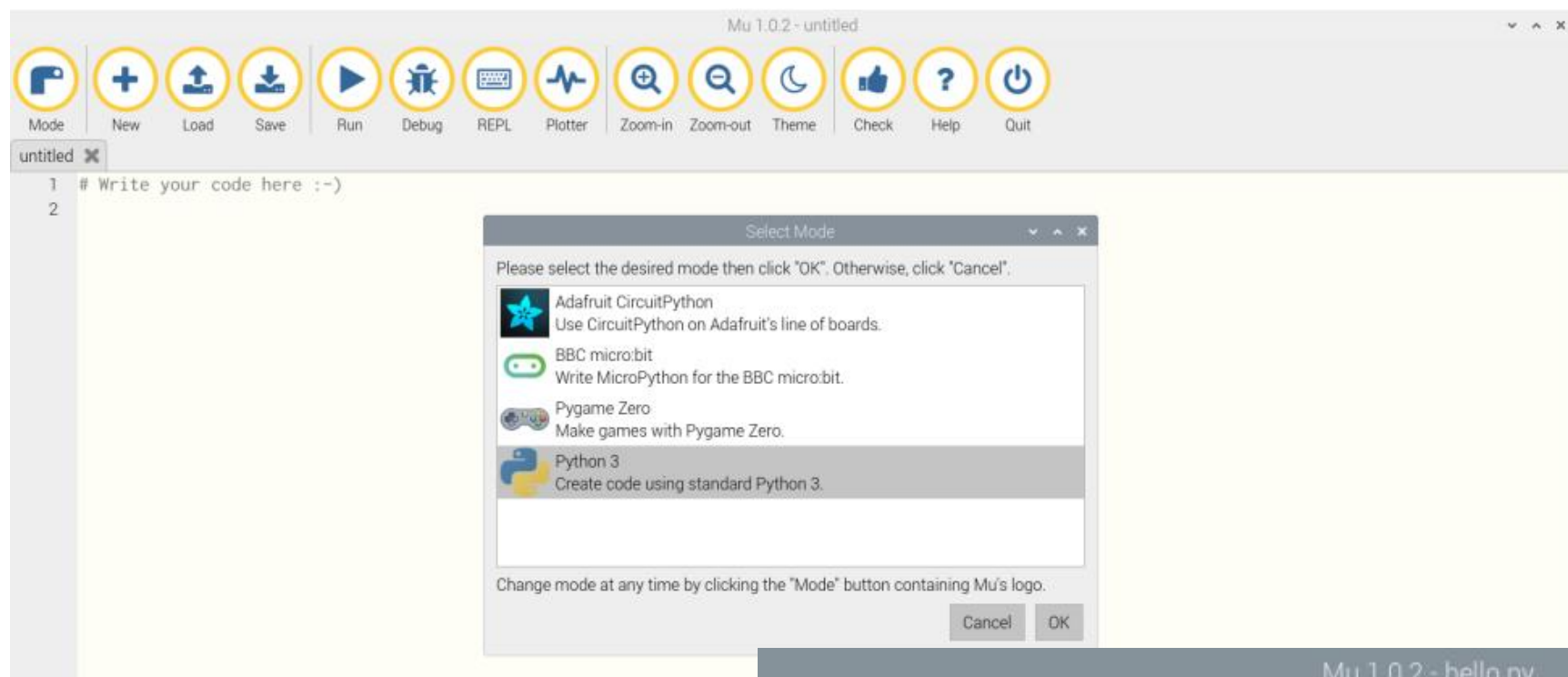
代码编辑器 (Code Editor)

Mu: a Python editor

<https://codewith.mu/>



打开 Mu 软件后，选择 Python 3 模式，点击 OK。



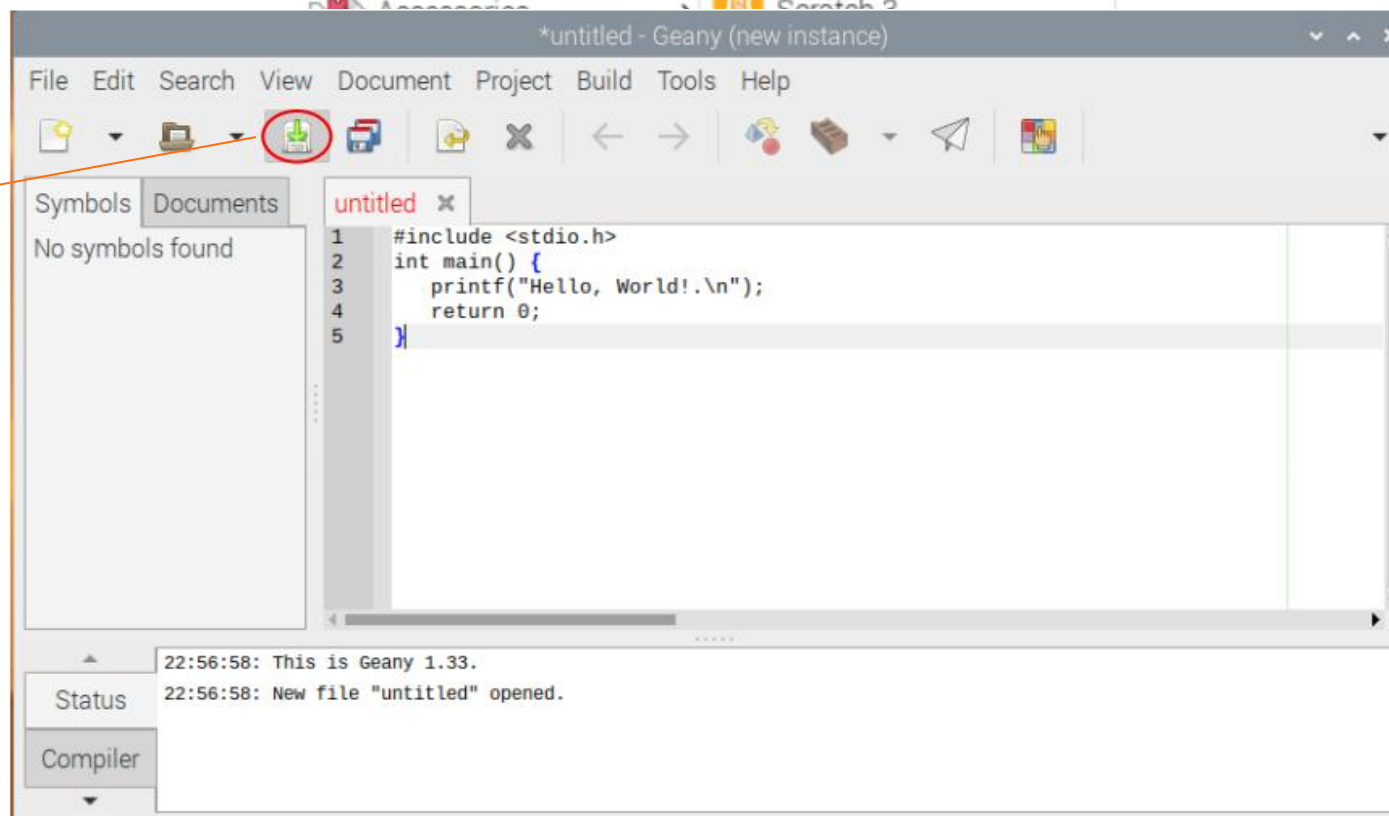
点击 Save 保存程序；点击 Check 检查程序（只要在程序未运行的情况下，都可以点击 Check 检查代码是否有错误）；点击 Run 运行程序；点击 Stop 停止运行程序



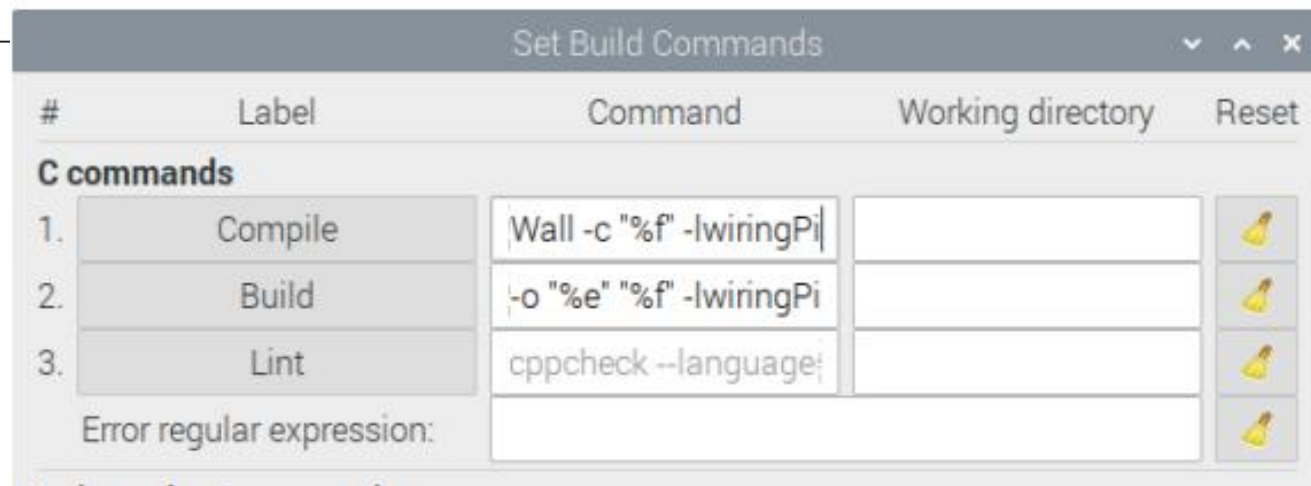
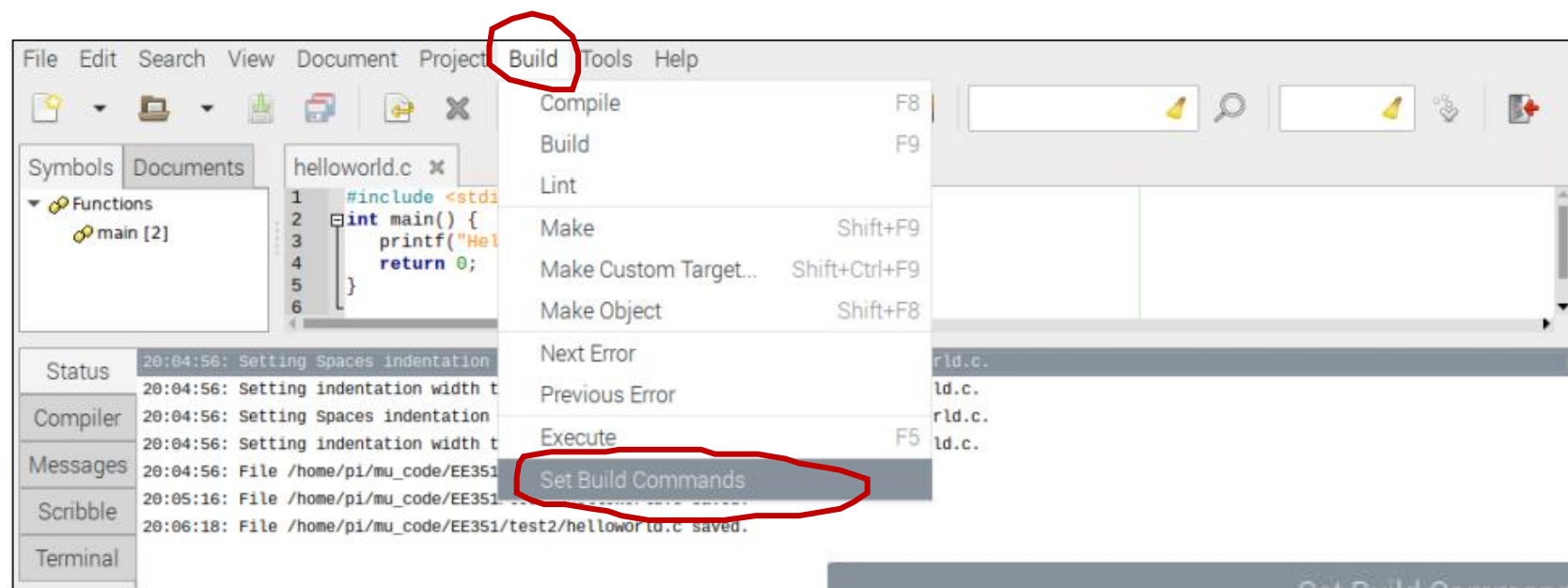
Geany,支持多种编程语言：
Python、C、C++、Java等



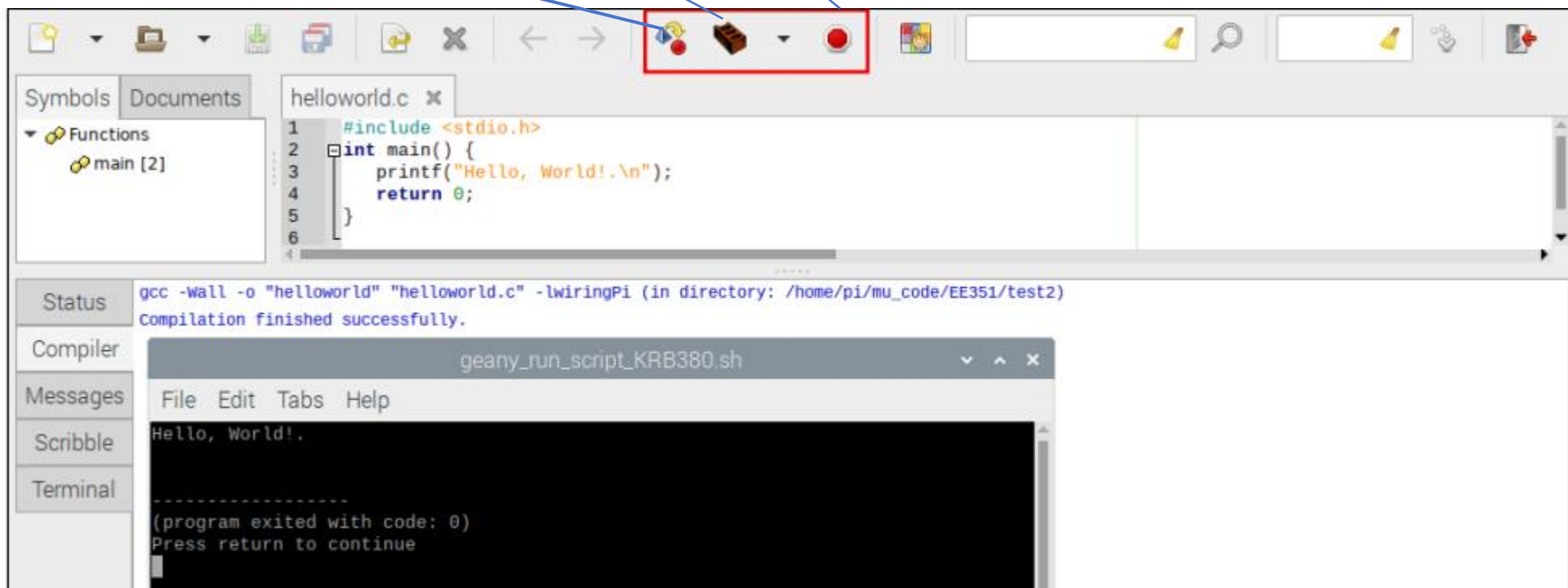
保存



如果使用 Geany 的 wiringPi 库编译 C/C++ 程序，需要进行配置，具体方法如下。点击上侧 “Build” 按钮，选择 Set Build Commands。在 Compile 和 Build 选框里最后面添加 “-lwiringPi”（空格LWIRINGPI）



三个按钮分别为编译文件、生成运行文件、执行。



文本编辑器 (Text Editor)

- 在树莓派（或 Linux 系统）中，nano 和 vi 都是命令行下的文本编辑器，用于在终端中编辑文件（如代码、配置文件等）。两者的设计理念和 Usage 方式差异较大。
 - nano 是一款轻量、简单、新手友好的文本编辑器，设计初衷是让用户能快速上手。它的特点是操作直观，底部会直接显示常用快捷键提示，不需要记忆复杂命令，适合初学者。
 - vi 是一款历史悠久、功能强大的文本编辑器，在所有 Linux 系统中几乎都预装。它的特点是模式化操作（需切换命令 / 插入模式），初期学习成本较高，但熟练后效率极高，适合编辑代码或配置文件。

文本编辑器	nano	vi	
基本用法	1. 打开 / 创建文件：nano 文件名（如 nano test.txt） 2. 编辑：直接输入文字（默认进入编辑模式） 3. 保存：Ctrl + O（回车确认文件名） 4. 退出：Ctrl + X	1. 打开 / 创建文件：vi 文件名（如 vi test.txt） 2. 进入编辑：命令模式下按 i（插入模式） 3. 保存退出：按 Esc 回命令模式，输入 :wq 并回车 4. 放弃退出：命令模式下输入 :q! 并回车	
常用快捷键	- Ctrl + O：保存文件 - Ctrl + X：退出编辑器 - Ctrl + K：剪切当前行 - Ctrl + U：粘贴剪切内容 - Ctrl + W：搜索文本 - Ctrl + \：替换文本 - Ctrl + C：显示光标位置	命令模式 - i：进入插入模式（在光标前编辑） - dd：删除当前行 - yy：复制当前行 - p：粘贴内容 - u：撤销操作 - Shift + zz：快速保存并退出 - h/j/k/l：左 / 下 / 上 / 右移动光标	末行模式 - :w：保存文件 - :q：退出（未保存时提示） - :q!：强制退出（不保存） - :wq：保存并退出 - :set nu：显示行号 - :set nonu：隐藏行号

Vi编辑器的3种模式

1.命令模式 (Command Mode)

- 启动 vi 后默认进入的模式，是所有操作的基础入口。
- 功能：用于移动光标、删除 / 复制文本、搜索替换、切换到其他模式等（不能直接输入文字）。
- 特征：屏幕底部无特殊提示，光标通常显示为闪烁的竖线或方块。

2.插入模式 (Insert Mode)

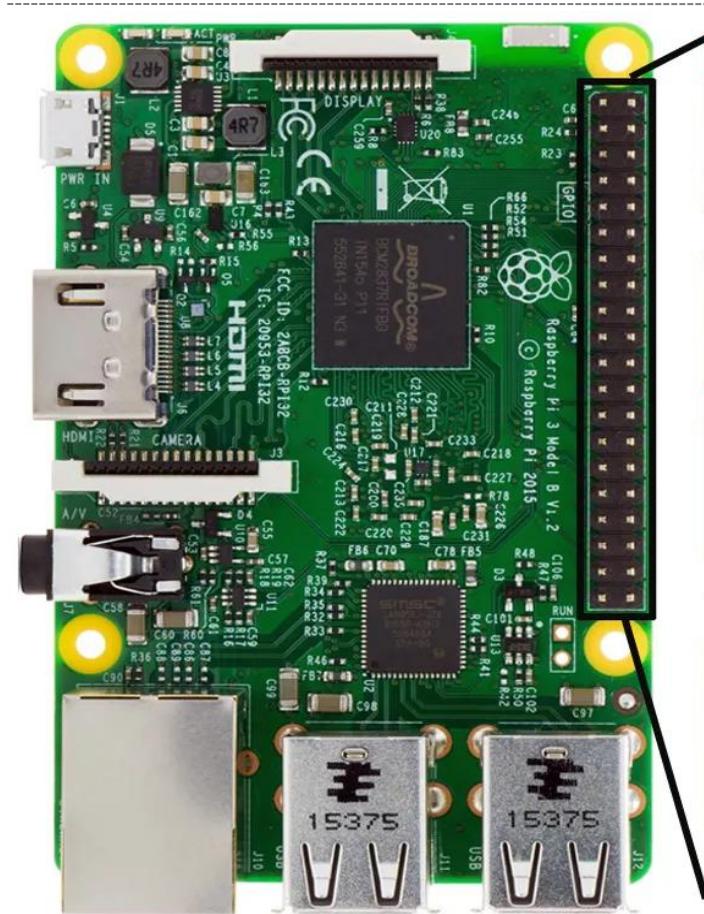
- 用于实际输入和编辑文本的模式。
- 功能：可直接键入文字，类似普通文本编辑器的编辑状态。
- 特征：屏幕底部会显示 -- INSERT -- 提示。

3.末行模式 (Last Line Mode)

- 用于执行保存、退出、全局设置等命令的模式。
- 功能：输入以冒号 : 开头的命令（如保存 :w、退出 :q 等），支持文件操作、查找替换、配置编辑器等。
- 特征：光标会移到屏幕底部的末行，等待输入命令（显示：提示符）。

•命令模式是“中转站”，所有模式切换都需经过它。插入模式用于“写”，末行模式用于“执行命令”，命令模式用于“控制和操作”。

硬件基础--GPIO管脚 (Hardware Basics—GPIO Pins)



A photograph of a Raspberry Pi 3 Model B V1.2. The 40-pin GPIO header is highlighted with a black box. A black arrow points from the 'Key' table below to the header. The board features a Broadcom BCM2837 processor, 1GB RAM, and various ports including USB, HDMI, and Ethernet.

	Pin No.	
3.3V	1	2
GPIO2	3	4
GPIO3	5	6
GPIO4	7	8
GND	9	10
GPIO17	11	12
GPIO27	13	14
GPIO22	15	16
3.3V	17	18
GPIO10	19	20
GPIO9	21	22
GPIO11	23	24
GND	25	26
DNC	27	28
GPIO5	29	30
GPIO6	31	32
GPIO13	33	34
GPIO19	35	36
GPIO26	37	38
GND	39	40

Key

Power +	UART	DNC
GND	SPI	
I ² C	GPIO	

3.3V PWR	1	2	5V PWR
I2C1 SDA	3	4	5V PWR
I2C1 SCL	5	6	GND
GPIO 4	7	8	TxD
GND	9	10	RxD
GPIO 17	11	12	GPIO 18
GPIO 27	13	14	GND
GPIO 22	15	16	GPIO 23
3.3V PWR	17	18	GPIO 24
SPI0 MOSI	19	20	GND
SPI0 MISO	21	22	GPIO 25
SPI0 SCLK	23	24	SPI0 CS0
GND	25	26	SPI0 CS1
Reserved	27	28	Reserved
GPIO 5	29	30	GND
GPIO 6	31	32	GPIO 12
GPIO 13	33	34	GND
GPIO 19	35	36	GPIO 16
GPIO 26	37	38	GPIO 20
GND	39	40	GPIO 21

引脚编码 (Pin Numbering)

- Raspberry Pi 目前存在的有三种编码规范，分别是：
- 1)BOARD:从左到右,从上到下:左边基数,右边偶数:1-40 (根据引脚的物理位置编号Header)
- 2)BCM:由 BCM2837SOC 指定的编号 (BCM GPIO) 编号侧重CPU寄存器,根据BCM2835的GPIO寄存器编号
- 3)wiringPi :由 C 语言 GPIO 库 wiringPi 指定的编号，编号侧重实现逻辑,把扩展GPIO端口从0开始编号,这种编号方便编程。

树莓派 40Pin 引脚对照表

wiringPi 编码	BCM 编码	功能名	物理引脚 BOARD编码		功能名	BCM 编码	wiringPi 编码
		3.3V	1	2	5V		
8	2	SDA.1	3	4	5V		
9	3	SCL.1	5	6	GND		
7	4	GPIO.7	7	8	TXD	14	15
		GND	9	10	RXD	15	16
0	17	GPIO.0	11	12	GPIO.1	18	1
2	27	GPIO.2	13	14	GND		
3	22	GPIO.3	15	16	GPIO.4	23	4
		3.3V	17	18	GPIO.5	24	5
12	10	MOSI	19	20	GND		
13	9	MISO	21	22	GPIO.6	25	6
14	11	SCLK	23	24	CE0	8	10
		GND	25	26	CE1	7	11
30	0	SDA.0	27	28	SCL.0	1	31
21	5	GPIO.21	29	30	GND		
22	6	GPIO.22	31	32	GPIO.26	12	26
23	13	GPIO.23	33	34	GND		
24	19	GPIO.24	35	36	GPIO.27	16	27
25	26	GPIO.25	37	38	GPIO.28	20	28
		GND	39	40	GPIO.29	21	29

表格由树莓派实验室绘制 <http://shumeipai.nxez.com>

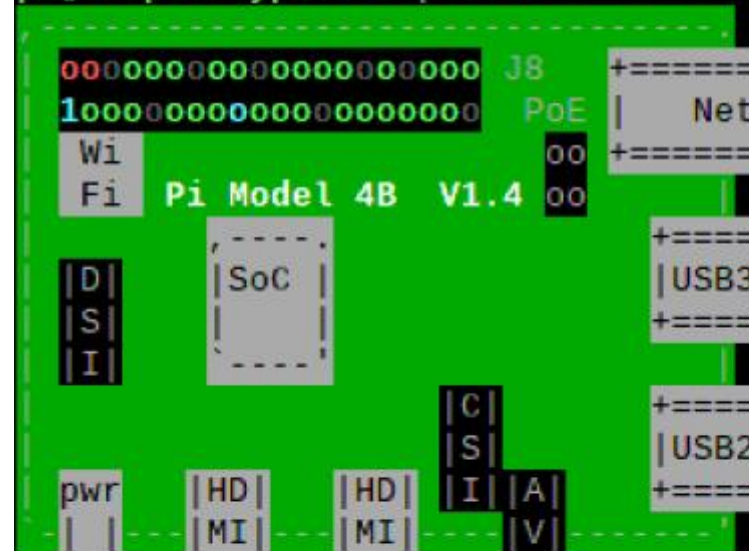
使用python控制树莓派
引脚时，使用BCM编码

使用C语言控制树莓派引脚时，
使用wiringpi编码

我们不需要记住这些，只需要知道如何在终端窗口中来查看这三种编码就可以

- **pinout**
- **gpio readall** (**gpio 命令是 WiringPi 库提供的命令行工具，必须安装 WiringPi 库后才能使用。**)


```
pi@raspberrypi:~ $ pinout
```



```
Revision      : c03114
SoC           : BCM2711
RAM           : 4096Mb
Storage       : MicroSD
USB ports     : 4 (excluding power)
Ethernet ports : 1
Wi-fi        : True
Bluetooth    : True
Camera ports (CSI) : 1
Display ports (DSI): 1
```

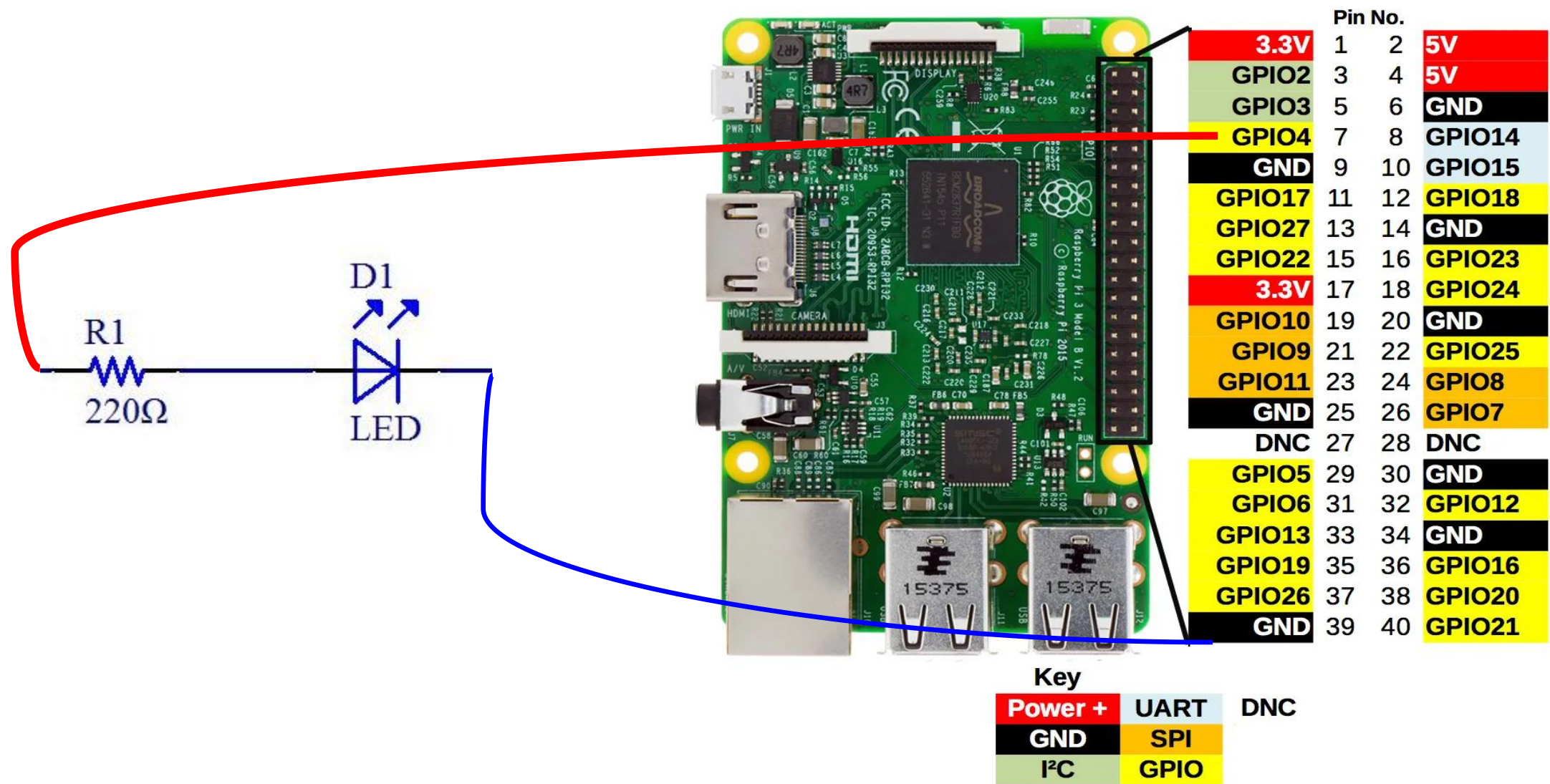
J8:

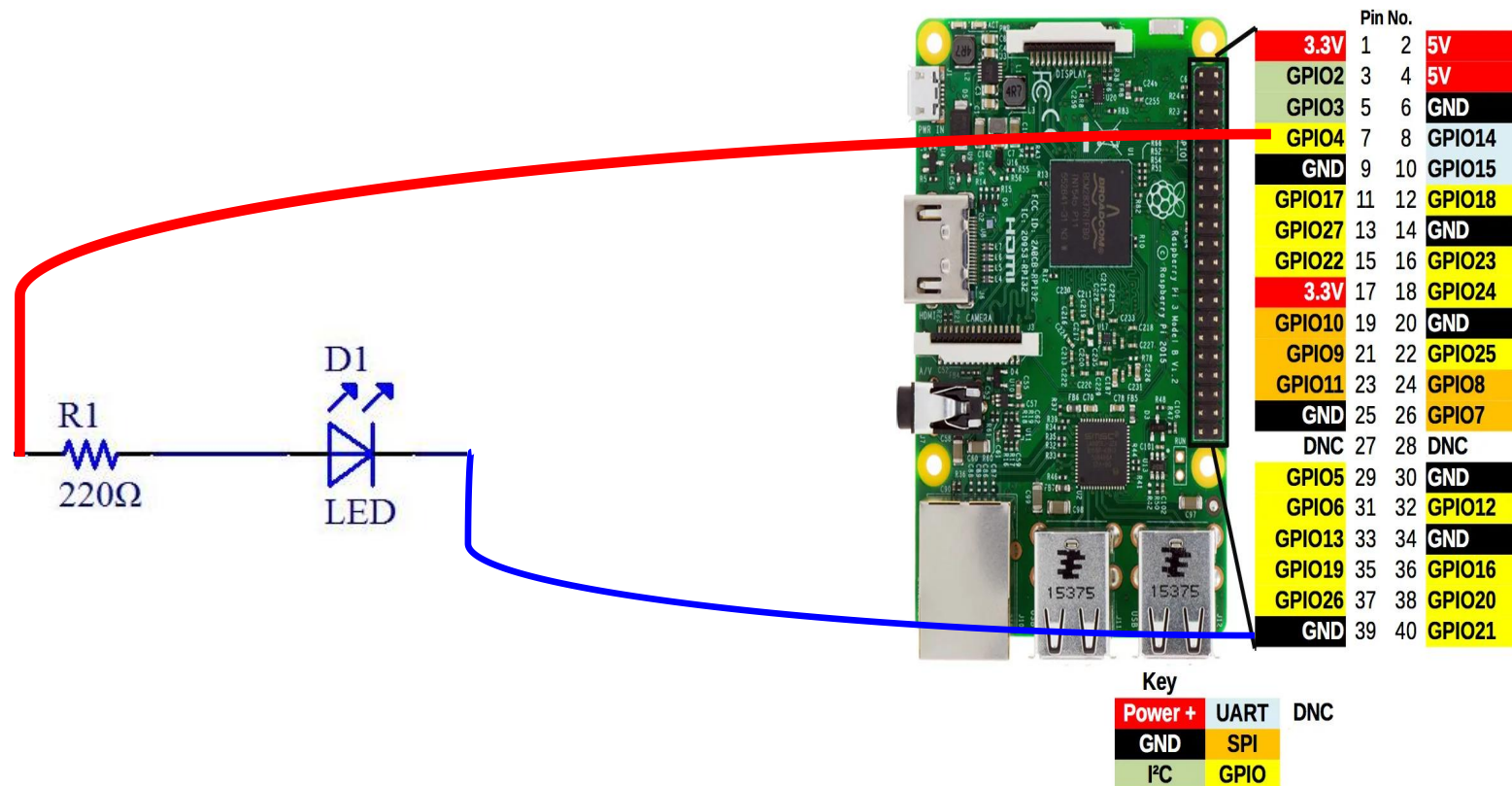
3V3	(1)	(2)	5V
GPI02	(3)	(4)	5V
GPI03	(5)	(6)	GND
GPI04	(7)	(8)	GPI014
GND	(9)	(10)	GPI015

```
pi@raspberrypi:~ $ gpio readall
```

Pi 4B											
BCM	wPi	Name	Mode	V	Physical	V	Mode	Name	wPi	BCM	
		3.3v			1	2		5v			
2	8	SDA.1	IN	1	3	4		5v			
3	9	SCL.1	IN	1	5	6		0v			
4	7	GPIO. 7	IN	1	7	8	1	IN	TxD	15	14
		0v			9	10	1	IN	RxD	16	15
17	0	GPIO. 0	IN	0	11	12	0	IN	GPIO. 1	1	18
27	2	GPIO. 2	IN	0	13	14		0v			
22	3	GPIO. 3	IN	0	15	16	0	IN	GPIO. 4	4	23
		3.3v			17	18	0	IN	GPIO. 5	5	24
10	12	MOSI	IN	0	19	20		0v			
9	13	MISO	IN	0	21	22	0	IN	GPIO. 6	6	25
11	14	SCLK	IN	0	23	24	1	IN	CE0	10	8
		0v			25	26	1	IN	CE1	11	7
0	30	SDA.0	IN	1	27	28	1	IN	SCL.0	31	1
5	21	GPIO.21	IN	1	29	30		0v			
6	22	GPIO.22	IN	1	31	32	0	IN	GPIO.26	26	12
13	23	GPIO.23	IN	0	33	34		0v			
19	24	GPIO.24	IN	0	35	36	0	IN	GPIO.27	27	16
26	25	GPIO.25	IN	0	37	38	0	IN	GPIO.28	28	20
		0v			39	40	0	IN	GPIO.29	29	21
BCM	wPi	Name	Mode	V	Physical	V	Mode	Name	wPi	BCM	
Pi 4B											

Lab1--双色灯实验 (Dual-Color LED)





```
gpio -g mode 4 out
```

#设置管脚为输出模式,-g表示是以BCM编码

如果要以wiringPi编码方式, 改写为: **gpio mode 7 out**

如果要以BOARD编码方式改写为: **gpio -l mode 7 out**

```
gpio -g read 4
```

#读取管脚当前状态

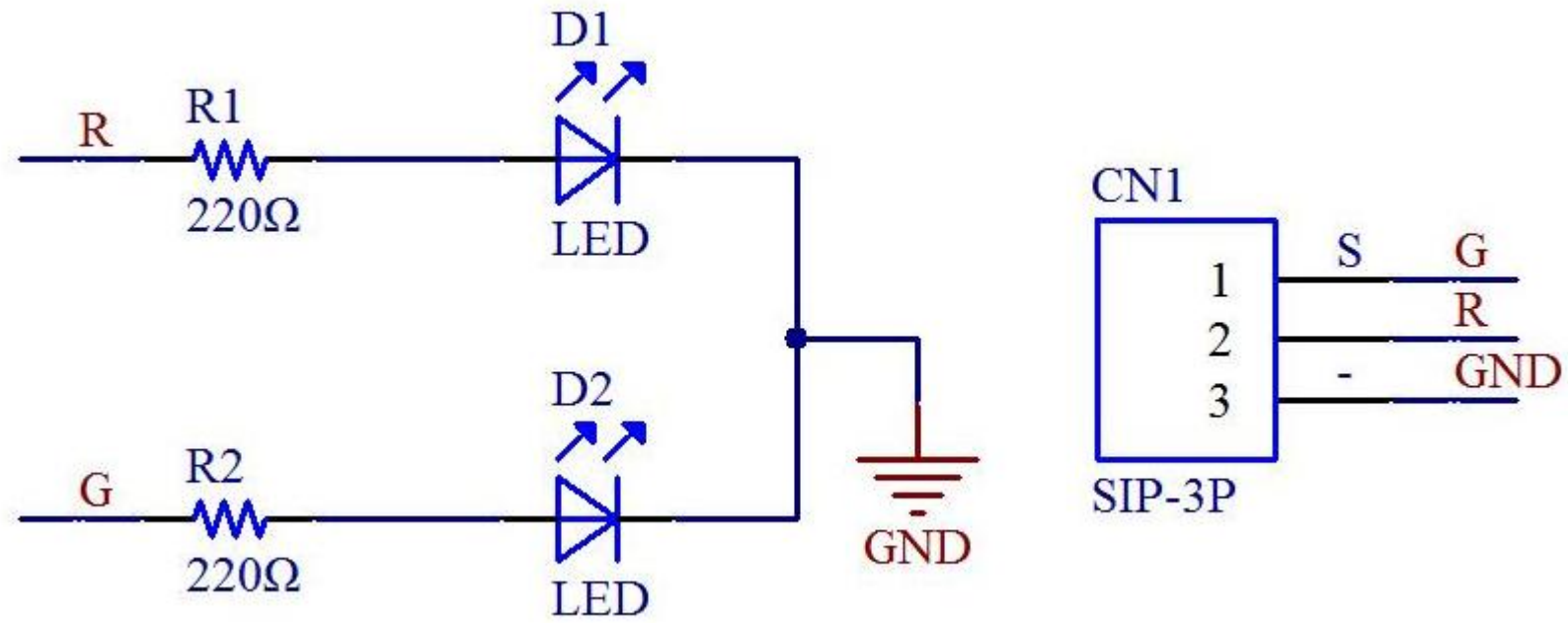
```
gpio -g write 4 1
```

#设置管脚为高电平

```
gpio -g write 4 0
```

#设置管脚为低电平

用gpio命令控制GPIO



使用RPI.GPIO库，python编码

RPI.GPIO模块是连接树莓派(GPIO)的Python库，可以帮助您使用Python编程控制外围设备。

- **引入模块**：在您的Python程序中引入RPI.GPIO库：`import RPi.GPIO as GPIO`
 - **设置GPIO模式**：在代码中设置GPIO为三种不同模式之一：输入（Input）、输出（Output）或板上定义（Board）。通常，在开始时将模式设置为板上定义（Board）模式：`GPIO.setmode(GPIO.BOARD)` 或 `GPIO.setmode(GPIO.BCM)`
- **配置GPIO引脚**：配置需要使用的GPIO引脚以及相应的方向（输入或输出）：`GPIO.setup(channel, GPIO.IN/OUT)`。这里的channel可以是物理编号（Physical Pin）或 BCM板上引脚编号，根据之前选的模式来。
- **控制GPIO引脚**：用适当的值来使GPIO引脚转换到所需状态，如果是输出状态则为High（高电平）或Low（低电平）；如果是输入状态，则检测当前的状态：`GPIO.output(channel, state)`和`GPIO.input(channel)`
- **清除引脚设置**：完成所有任务之后，必须释放GPIO引脚，清除之前的设置以便它们能够还原为其初始状态：`GPIO.cleanup()`

<https://sourceforge.net/p/raspberry-gpio-python/wiki/BasicUsage/>

RPI.GPIO库里面的函数怎么查

- RPI.GPIO库是用于控制和监控你的Raspberry Pi上GPIO（通用输入/输出）口的Python库。你可以使用help函数或者直接查看官方文档来查看库中的函数和它们的使用方法。
- 以下是一些查看RPI.GPIO库函数的方法：
 - 使用Python的help函数：
 - `import RPi.GPIO as GPIO`
 - `help(GPIO)`
 - 这将打印出GPIO模块的文档字符串和所有函数列表。
 - 查看官方文档：
 - 你可以在Raspberry Pi的官方文档中找到Python GPIO库的详细信息和示例代码：
 - <https://www.raspberrypi.org/documentation/computers/python-computers.html>
 - <https://sourceforge.net/p/raspberry-gpio-python/wiki/Home/>
 - 使用dir函数列出所有可用的函数：
 - `import RPi.GPIO as GPIO`
 - `print(dir(GPIO))`
 - 这将列出所有可以使用的函数和类。
 - 使用Python的help函数查看特定函数的用法：
 - `import RPi.GPIO as GPIO`
 - `help(GPIO.setmode)`
 - 这将打印出setmode函数的文档字符串，包括它的功能、参数列表和返回值。

```
import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BOARD)
GPIO.setup(7, GPIO.OUT)
try:
    while True:
        GPIO.output(7, GPIO.HIGH)
        print("LED ON")
        time.sleep(1)
        GPIO.output(7, GPIO.LOW)
        print("LED OFF")
        time.sleep(1)
except KeyboardInterrupt:
    print("用户终止程序")
finally:
    GPIO.cleanup() # 关键：确保退出时释放
    print("程序退出，GPIO已清理")
```

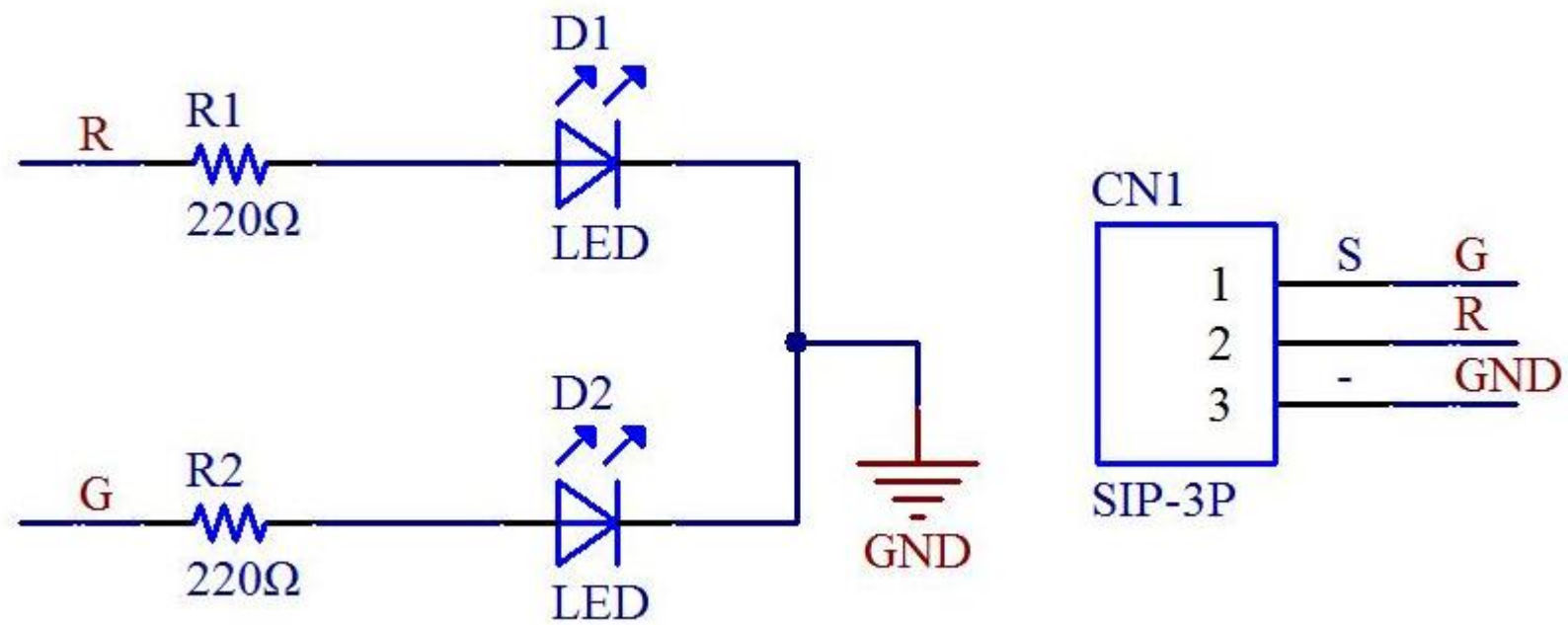
```
import RPi.GPIO as GPIO
import time
led=7
GPIO.setmode(GPIO.BOARD)
GPIO.setup(led, GPIO.OUT)
try:
    while True:
        GPIO.output(led, GPIO.HIGH)
        print("LED ON")
        time.sleep(1)
        GPIO.output(led, GPIO.LOW)
        print("LED OFF")
        time.sleep(1)
except KeyboardInterrupt:
    print("用户终止程序")
finally:
    GPIO.cleanup() # 关键：确保退出时释放
    print("程序退出，GPIO已清理")
```

```
led=(7,8)
GPIO.setmode(GPIO.BOARD)
GPIO.setup(led, GPIO.OUT)
```

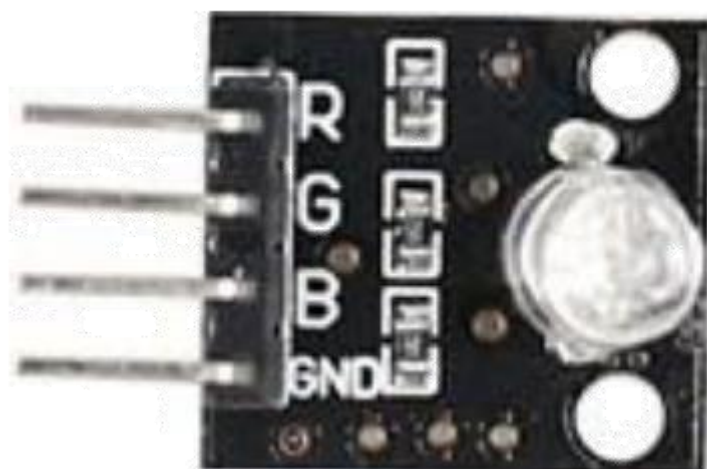
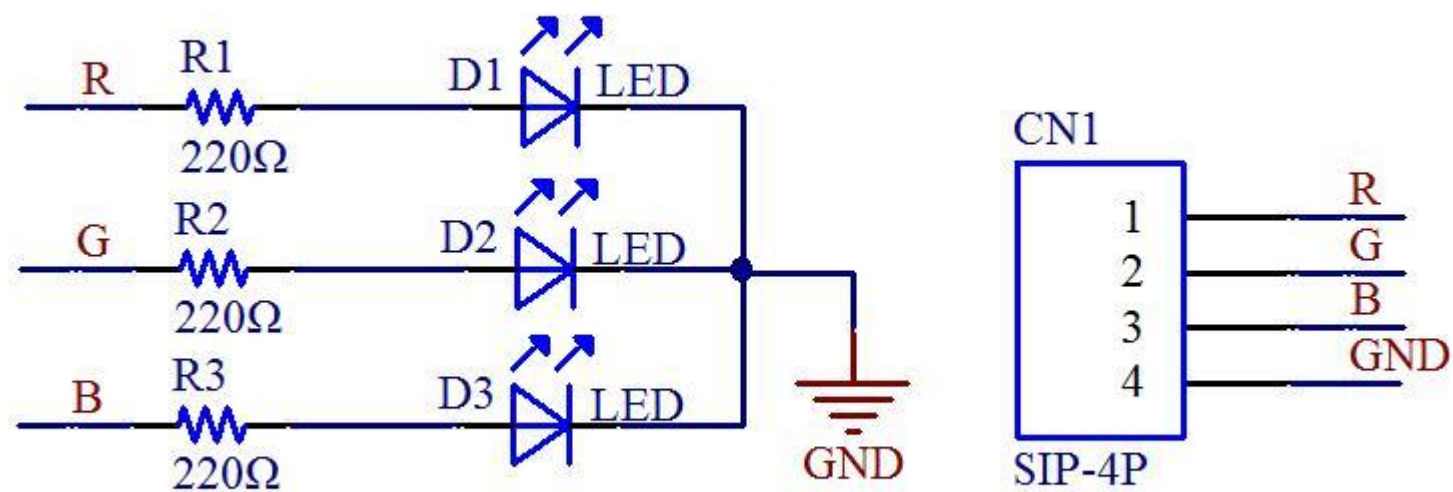
使用wiringPi库，C编码

- 使用带有 Geany 的 wiringPi 库编译 C/C++程序，需要进行配置，具体方法如下。点击上侧 “Build” 按钮，选择 Set Build Commands。
- 在 Compile 和 Build 选框里最后面添加 “-lwiringPi” （空格 LWIRINGPI ）
- 在终端窗口中按下CTRL+C停止程序运行。

```
#include <stdio.h>
#include <wiringPi.h>
#define LED_PIN 7 // 使用wiringPi引脚编号
int main() {
    // 初始化wiringPi库
    if (wiringPiSetup() == -1) {
        printf("Setup wiringPi failed!");
        return 1;
    }
    // 设置LED引脚模式为输出
    pinMode(LED_PIN, OUTPUT);
    // 循环控制LED开和关
    for (;;) {
        digitalWrite(LED_PIN, HIGH); // 开灯
        delay(500);                  // 等待500毫秒
        digitalWrite(LED_PIN, LOW);  // 关灯
        delay(500);                  // 等待500毫秒
    }
    return 0;
}
```



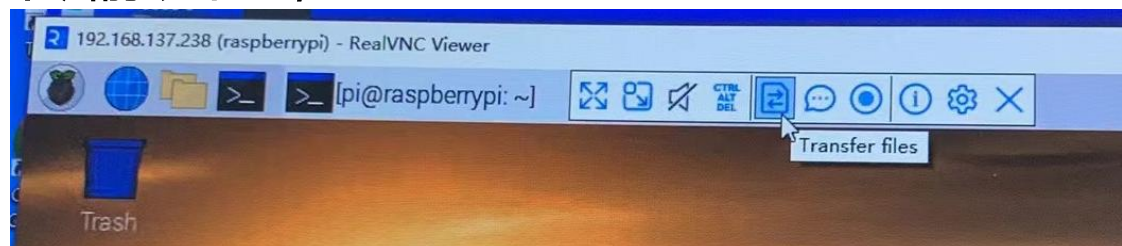




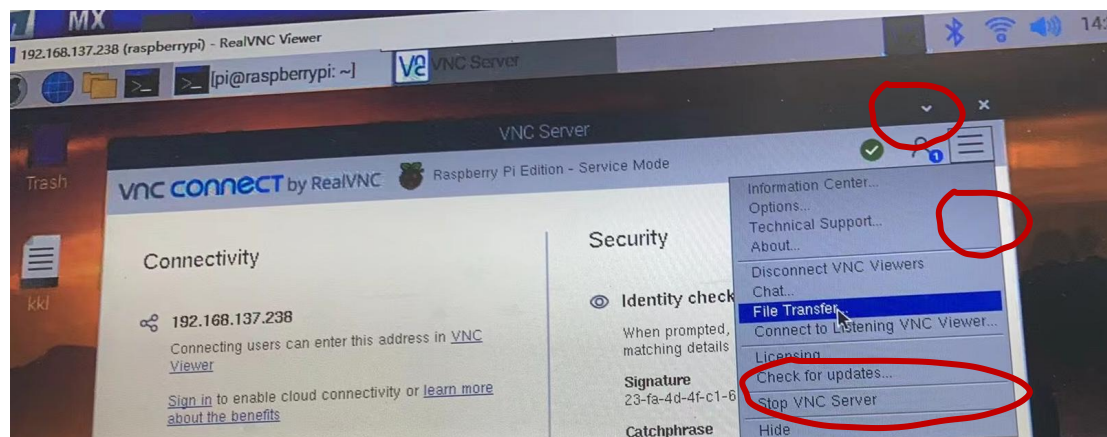
基础--文件传输 (file transfer)

- 1：U盘传输，插在树莓派上；
- 2：VNC viewer自带了文件传输功能；

电脑到树莓派



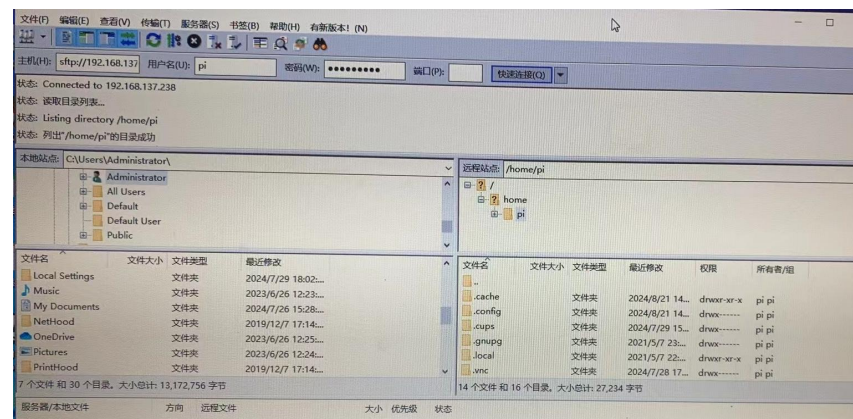
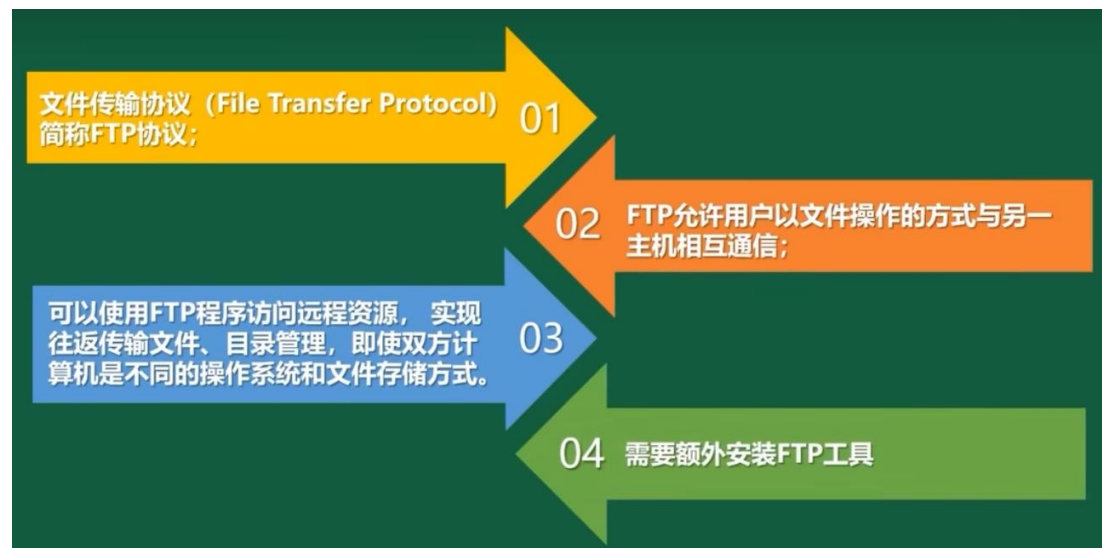
树莓派到电脑



如果图标为灰色，按照该网页设置一下：

https://yebd1h.smartapps.cn/pages/blog/index?blogId=147046479&_swebfr=1&_swebFromHost=baiduboxapp

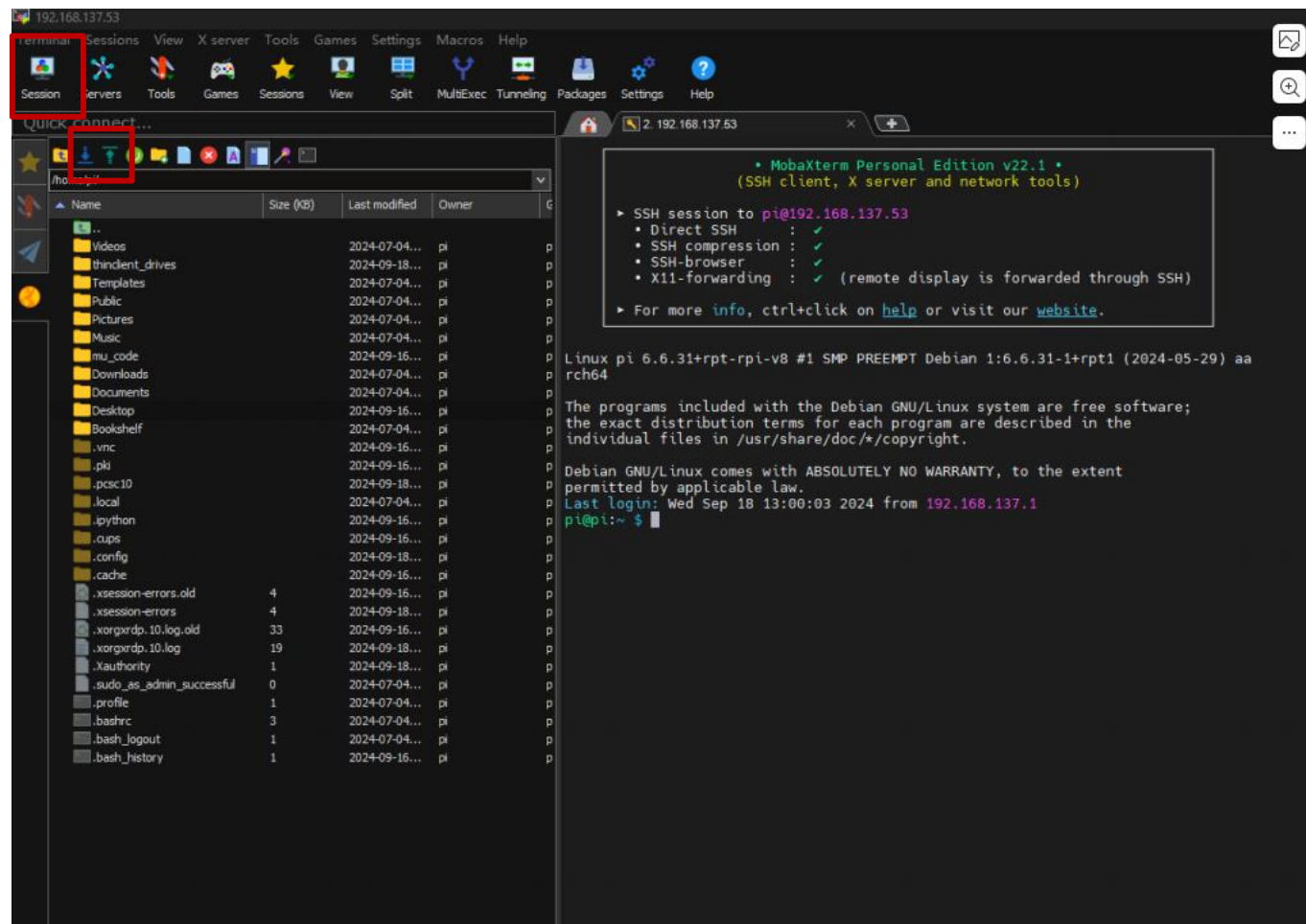
• 3 , FTP文件传输协议



4 , MobaXterm----全程终端神器



MobaXterm

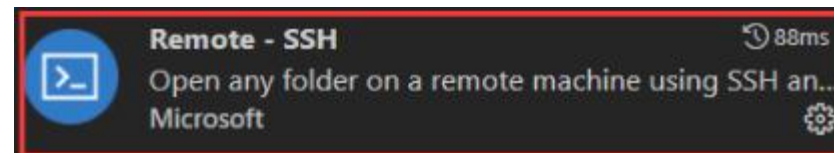


5 , vscode远程连接树莓派

https://blog.csdn.net/weixin_44562141/article/details/120299724

配置vscode

- 打开vscode , 在扩展里搜索安装 remote ssh插件



- 插件安装完成之后 , 左边菜单会多出一个“远程资源管理器”
- 创建远程配置 (按照提示 , 加入树莓派)
- 右键新加入的主机 , 选择linux , 输入密码即可完成连接

